Approximate counting requires a large untrusted quantum witness or quadratically many membership queries

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Editorial note. Editable transcription of corresponding published CC BY3.0 Unported QMA problem/proof; nonexclusive nativev3 is private aid, not a public original. Selected public files exclude all unselected raw ranges/comments. FullSection2 andSection3 preserved, including original author constants/probability/parameter expressions without mathematical correction or added assumptions. Original restatable theorem numbers, equation offset and Figure3 restored; author TikZ figure retained as editable body with explicit actual package dependencies. Standard prior facts cited rather than recursively reproved. Native malformed optional bibliography label avoided by using formal published bibliographic text with original numbers. All production/validation solelyYupeng; root owns compilation.

Allowed standard prerequisites (editorial).

- Quantum query algorithms and QMA amplification; approximation theory and symmetric polynomial symmetrization. Full originalSection2 definitions/statements, source-local symmetrization proofs and guessing lemma retained; standard cited amplification is not recursively reproved

Approximate counting and the untrusted witness model

Problem 1 (Approximate Counting)**.** *In the* $\mathsf{ApxCount}_{N,w}$ *problem, our goal is to decide whether a nonempty set* $S \subseteq [N]$ *satisfies* $|S| \geq 2w$ *(YES) or* $|S| \leq w$ *(NO), promised that one of these is the case.*

In the query model, the algorithm is given a membership oracle for S: one that, for any $i \in [N]$, returns whether $i \in S$. In a QMA protocol, a skeptical verifier (Arthur) receives a quantum witness state $|\psi\rangle$ from an all-powerful but untrustworthy prover (Merlin), in support of the claim that $f(x) = 1$. Arthur then needs to verify $|\psi\rangle$, via some algorithm that satisfies the twin properties of completeness and soundness. That is, if $f(x) = 1$, then there must exist some $|\psi\rangle$ that causes Arthur to accept with high probability, while if $f(x) = 0$, then every $|\psi\rangle$ must cause Arthur to reject with high probability. We call such a protocol a QMA protocol for computing f.

Theorem 2. *Consider a* QMA *protocol that solves* $\mathsf{ApxCount}_{N,w}$. *If the protocol receives a quantum witness of length* m, *and makes* T *queries to the membership oracle for* S, *then either* $m = \Omega(w)$ *or* $T = \Omega\big(\sqrt{N/w}\big)$.

This lower bound proved in Section 3.2 resolves the QMA complexity of $\mathsf{ApxCount}_{N,w}$, as (up to a $\log N$ factor) it matches the cost of two trivial QMA protocols. In the first, Merlin sends $2w$ items claimed to be in S, and Arthur picks a constant number of the items at random and confirms they are all in S with one membership query each. This protocol has witness length $m = O(w \log N)$ (the number of bits needed to specify $2w$ elements out of N) and $T = O(1)$. In the second protocol, Merlin does nothing, and Arthur solves the problem with $T = O\big(\sqrt{N/w}\big)$ quantum queries.

Theorem 3. *Consider an* SBQP *algorithm for* $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ *that makes* T *queries to membership oracles for the two instances of* $\mathsf{ApxCount}_{N,w}$. *Then* $T = \Omega\left(\min\big\{w, \sqrt{N/w}\big\}\right)$.

2 Preliminaries

In this section we introduce some definitions and known facts about polynomials and complexity classes.

2.1 Approximation theory

We will use several results from approximation theory, each of which has previously been used (in some form) in other applications of the polynomial method to quantum lower bounds. We start with the basic inequality of A.A. Markov [35].

Lemma 6 (Markov)**.** *Let* p *be a real polynomial, and suppose that*

$$\max_{x,y \in [a,b]} |p(x) - p(y)| \leq H. \tag{5}$$

Then for all $x \in [a,b]$, *we have*

$$|p'(x)| \leq \frac{H}{b-a} \deg(p)^2, \tag{6}$$

where $p'(x)$ *is the derivative of* p *at* x.

We'll also need a bound that was explicitly stated by Paturi [43], and which amounts to the fact that, among all degree-d polynomials that are bounded within a given range, the Chebyshev polynomials have the fastest growth outside that range.

Lemma 7 (Paturi)**.** *Let* p *be a real polynomial, and suppose that* $|p(x)| \leq 1$ *for all* $|x| \leq 1$. *Then for all* $x \leq 1 + \mu$, *we have*

$$|p(x)| \leq \exp\left(2 \deg(p) \sqrt{2\mu + \mu^2}\right). \tag{7}$$

We now state a useful corollary of Lemma 7, which says (in effect) that slightly shrinking the domain of a low-degree real polynomial can only modestly shrink its range.

Corollary 8. *Let p be a real polynomial of degree d, and suppose that*

$$\max_{x,y\in[a,b]} |p(x) - p(y)| \geq H. \tag{8}$$

Let $\varepsilon \leq \frac{1}{100d^2}$ and $a' := a + \varepsilon(b-a)$. Then

$$\max_{x,y\in[a',b]} |p(x) - p(y)| \geq \frac{H}{2}. \tag{9}$$

Proof. Suppose by contradiction that

$$|p(x) - p(y)| < \frac{H}{2} \tag{10}$$

for all $x, y \in [a', b]$. By affine shifts, we can assume without loss of generality that $|p(x)| < \frac{H}{4}$ for all $x \in [a', b]$. Then by Lemma 7, for all $x \in [a, b]$ we have

$$|p(x)| < \frac{H}{4} \cdot \exp\left(2d\sqrt{2\left(\frac{1}{1-\varepsilon} - 1\right) + \left(\frac{1}{1-\varepsilon} - 1\right)^2}\right) \leq \frac{H}{2}. \tag{11}$$

But this violates the hypothesis. □

We will also need a bound that relates the range of a low-degree polynomial on a discrete set of points to its range on a continuous interval. The following lemma generalizes a result due to Ehlich and Zeller [22] and Rivlin and Cheney [46], who were interested only in the case where the discrete points are evenly spaced.

Lemma 9. *Let p be a real polynomial of degree at most $\sqrt{k}$, and let $0 = z_1 < \cdots < z_M = k$ be a list of points such that $z_{i+1} - z_i \leq 1$ for all i (the simplest example being the integers $0, \ldots, k$). Suppose that*

$$\max_{x,y\in[0,k]} |p(x) - p(y)| \geq H. \tag{12}$$

Then

$$\max_{i,j} |p(z_i) - p(z_j)| \geq \frac{H}{2}. \tag{13}$$

Proof. Suppose by contradiction that

$$|p(z_i) - p(z_j)| < \frac{H}{2} \tag{14}$$

for all i, j. By affine shifts, we can assume without loss of generality that $|p(z_i)| < \frac{H}{4}$ for all i. Let

$$c := \max_{x\in[0,k]} \frac{|p(x)|}{H/4}. \tag{15}$$

If $c \leq 1$, then the hypothesis clearly fails, so assume $c > 1$. Suppose that the maximum, $|p(x)| = \frac{cH}{4}$, is achieved between z_i and z_{i+1}. Then by basic calculus, there exists an $x^* \in [z_i, z_{i+1}]$ such that

$$|p'(x^*)| > \frac{2(c-1)}{z_{i+1} - z_i} \cdot \frac{H}{4} \geq \frac{(c-1)H}{2}. \tag{16}$$

So by Lemma 6,

$$\frac{(c-1)\,H}{2} < \frac{cH/4}{k}\deg(p)^2. \tag{17}$$

Solving for c, we find

$$c < \frac{2k}{2k-\deg(p)^2} \leq 2. \tag{18}$$

But if $c < 2$, then $\max_{x\in[0,k]} |p(x)| < \frac{H}{2}$, which violates the hypothesis. □

We also use a related inequality due to Coppersmith and Rivlin [20] that bounds a polynomial on a continuous interval in terms of a bound on a discrete set of points, but now with the weaker assumption that the degree is at most k, rather than $\sqrt{k}$. This gives a substantially weaker bound.

Lemma 10 (Coppersmith and Rivlin)**.** *Let p be a real polynomial of degree at most k, and suppose that $|p(x)| \leq 1$ for all integers $x \in \{0,1,\ldots,k\}$. Then there exist universal constants a, b such that for all $x \in [0,k]$, we have*

$$|p(x)| \leq a \cdot \exp\left(b\,\deg(p)^2/k\right). \tag{19}$$

2.2 Symmetric polynomials

Univariate symmetrizations. Our starting point is the well-known *symmetrization lemma* of Minsky and Papert [39] (see also Beals et al. [8] for its application to quantum query complexity), by which we can often reduce questions about multivariate polynomials to questions about univariate ones.

Lemma 11 (Minsky–Papert symmetrization)**.** *Let $p : \{0,1\}^N \to \mathbb{R}$ be a real multilinear polynomial of degree d, and let $q : \{0,1,\ldots,N\} \to \mathbb{R}$ be defined as*

$$q(k) := \mathbb{E}_{|X|=k}[p(X)]. \tag{20}$$

Then q can be written as a real polynomial in k of degree at most d.

We now introduce a different, lesser known notion of symmetrization, which we call the *erase-all-subscripts* symmetrization for reasons to be explained shortly. This symmetrization previously appeared in [51] under the name "linearization," and it is also equivalent to the noise operator used in analysis of Boolean functions [42, Definition 2.46].

Lemma 12 (Erase-all-subscripts symmetrization)**.** *Let $p : \{0,1\}^N \to \mathbb{R}$ be a real multilinear polynomial of degree d, and for any real number $k \in [0,1]$, let M_k denote the distribution over $\{0,1\}^N$, wherein each coordinate is selected independently to be 1 with probability k. Let $q : [0,1] \to \mathbb{R}$ be defined as*

$$q(k) := \mathbb{E}_{X\sim M_k}[p(X)]. \tag{21}$$

Then q can be written as a real polynomial in k of degree at most d.

Proof. (see, for example, [47, Proof of Theorem 3]). Given the multivariate polynomial expansion of p, we can obtain q easily just by "erasing all the subscripts in each variable". For example, if $p(x_1,x_2,x_3) = 2x_1x_2 + x_2x_3 + x_2$, we replace every x_i with k to obtain $q(k) = 2k\cdot k + k\cdot k + k = 3k^2 + k$. This follows from linearity of expectation along with the fact that M_k is defined to be the product distribution wherein each coordinate has expected value k. □

We highlight the following key difference between Minsky–Papert symmetrization and the erase-all-subscripts symmetrization. Let $p : \{0,1\}^N \to [0,1]$ be a real multivariate polynomial whose evaluations at Boolean inputs are in $[0,1]$, i.e., for all $x \in \{0,1\}^n$, we have $p(x) \in [0,1]$. If q is the erase-all-subscripts symmetrization of p, then q takes values in $[0,1]$ at all *real-valued* inputs in $[0,1]$: $q(k) \in [0,1]$ for all $k \in [0,1]$. If q is the Minsky–Papert symmetrization of p, then it is only guaranteed to take values in $[0,1]$ at *integer-valued* inputs in $[0,N]$, i.e., $q(k) \in [0,1]$ is only guaranteed to hold at $k \in \{0,1,\ldots,N\}$. This is the main reason we use erase-all-subscripts symmetrization in this work.

Bivariate symmetrizations. In this paper, it will be convenient to consider bivariate versions of both Minsky–Papert and erase-all-subscripts symmetrization, and their applications to oracle separations. To this end, define $X \in \{0,1\}^N$, the "characteristic string" of the set $S \subseteq [N]$, by $x_i = 1$ if $i \in S$ and $x_i = 0$ otherwise. Let $\mathcal{O}_S$ denote the unitary that performs a membership query to S, defined as

$$\mathcal{O}_S |i\rangle |b\rangle = (1 - 2bx_i) |i\rangle |b\rangle \tag{22}$$

for any index $i \in [N]$ and bit $b \in \{0,1\}$.

Because we study oracle intersection problems, it is often convenient to think of an algorithm as having access to *two* oracles, wherein the first bit in the oracle register selects the choice of oracle. As a consequence, we need a slight generalization of a now well-established fact in quantum complexity: that the acceptance probability of a quantum algorithm with an oracle can be expressed as a polynomial in the bits of the oracle string.

Lemma 13 (Symmetrization with two oracles)**.** *Let $Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}}$ be a quantum algorithm that makes T queries to a pair of membership oracles for sets $S_0, S_1 \subseteq [N]$. Let D_μ denote the distribution over subsets of $[N]$ wherein each element is selected independently with probability $\frac{\mu}{N}$. Then there exist bivariate real polynomials $q(s,t)$ and $p(x,y)$ of degree at most $2T$ satisfying:*

$$\textit{for all real numbers } s,t \in [0,N], \quad q(s,t) = \mathbb{E}_{\substack{S_0 \sim D_s, \\ S_1 \sim D_t}} \left[\Pr[Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}} \textit{ accepts}]\right], \textit{ and}$$

$$\textit{for all integers } x,y \in \{0,1,\ldots,N\}, \quad p(x,y) = \mathbb{E}_{\substack{|S_0|=x, \\ |S_1|=y}} \left[\Pr[Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}} \textit{ accepts}]\right].$$

Proof. Take $X = X_0|X_1$ to be the concatenation of the characteristic strings of the two oracles, and let $S \subseteq [2N]$ be such that X is the characteristic string of S. Then, Lemma 4.2 of Beals et al. [8] tells us that there is a real multilinear polynomial $r(X)$ of degree at most $2T$ in the bits of X such that $r(X) = \Pr[Q^{\mathcal{O}_S} \text{ accepts}]$.

Observe that r has a meaningful probabilistic interpretation over arbitrary inputs in $[0,1]$. A vector $X \in [0,1]^{2N}$ of probabilities corresponds to a distribution over $\{0,1\}^{2N}$ wherein each bit is chosen from a Bernoulli distribution with the corresponding probability. Because r is multilinear, r in fact computes the expectation of the acceptance probability over this distribution. In particular, the polynomial

$$q(s,t) = r\Bigg(\underbrace{\frac{s}{N},\ldots,\frac{s}{N}}_{N \text{ times}},\underbrace{\frac{t}{N},\ldots,\frac{t}{N}}_{N \text{ times}}\Bigg) = \mathbb{E}_{\substack{S_0 \sim D_s, \\ S_1 \sim D_t}} \left[\Pr[Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}} \text{ accepts}]\right] \tag{23}$$

corresponds to selecting $S_0 \sim D_s$ and $S_1 \sim D_t$. The total degree of q is obviously at most the degree of r, by the same reasoning as in the proof of Lemma 12.

To construct p, we apply the symmetrization lemma of Minsky and Papert [39] to symmetrize r, first with respect to X_0, then with respect to X_1:

$$p_0(x, X_1) = \mathbb{E}_{|S_0|=x}\, r(X_0, X_1) = \mathbb{E}_{|S_0|=x}\left[\Pr[Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}} \text{ accepts}]\right] \tag{24}$$

$$p(x, y) = \mathbb{E}_{|S_1|=y}\, p_0(x, X_1) = \mathbb{E}_{\substack{|S_0|=x,\\ |S_1|=y}}\left[\Pr[Q^{\mathcal{O}_{S_0},\mathcal{O}_{S_1}} \text{ accepts}]\right] \tag{25}$$

The degree of p is at most the degree of r, due to Lemma 11. □

We remark that, as a consequence of their definitions in Lemma 13, p and q satisfy:

$$q(s, t) = \mathbb{E}\left[p(X, Y)\right], \tag{26}$$

where X and Y are drawn from N-trial binomial distributions with means s and t, respectively.

Symmetric Laurent polynomials. Finally, we state a useful fact about Laurent polynomials:

Lemma 14 (Symmetric Laurent polynomials)**.** *Let $\ell(x)$ be a real Laurent polynomial of positive and negative degree d that satisfies $\ell(x) = \ell(1/x)$. Then there exists a (ordinary) real polynomial q of degree d such that $\ell(x) = q(x + 1/x)$.*

Proof. $\ell(x) = \ell(1/x)$ implies that the coefficients of the x^i and x^{-i} terms are equal for all i, as otherwise $\ell(x) - \ell(1/x)$ would not equal the zero polynomial. Thus, we may write $\ell(x) = \sum_{i=0}^{d} a_i \cdot (x^i + x^{-i})$ for some coefficients a_i. So, it suffices to show that $x^i + x^{-i}$ can be expressed as a polynomial in $x + 1/x$ for all $0 \leq i \leq d$.

We prove by induction on i. The case $i = 0$ corresponds to constant polynomials. For $i > 0$, by the binomial theorem, observe that $(x + 1/x)^i = x^i + x^{-i} + r(x)$ where r is a degree $i - 1$ real Laurent polynomial satisfying $r(x) = r(1/x)$. By the induction assumption, r can be expressed as a polynomial in $x + 1/x$, so we have $x^i + x^{-i} = (x + 1/x)^i - r(x)$ is expressed as a polynomial in $x + 1/x$. □

2.3 Complexity classes

Definition 15. *The complexity class* QMA *consists of the languages L for which there exists a quantum polynomial time verifier V with the following properties:*

1. *Completeness: if $x \in L$, then there exists a quantum witness state $|\psi\rangle$ on* $\text{poly}(|x|)$ *qubits such that* $\Pr\left[V(x, |\psi\rangle) \text{ accepts}\right] \geq \frac{2}{3}$.
2. *Soundness: if $x \notin L$, then for any quantum witness state $|\psi\rangle$ on* $\text{poly}(|x|)$ *qubits,* $\Pr\left[V(x, |\psi\rangle) \text{ accepts}\right] \leq \frac{1}{3}$.

A quantum verifier that satisfies the above promise for a particular language will be referred to as a QMA verifier or QMA protocol throughout.

Though SBP and SBQP can be defined in terms of counting complexity functions, for our purposes it is easier to work with the following equivalent definitions (see Böhler et al. [11]):

Definition 16. *The complexity class* SBP *consists of the languages L for which there exists a probabilistic polynomial time algorithm M and a polynomial σ with the following properties:*

1. *If $x \in L$, then* $\Pr[M(x)\ accepts] \geq 2^{-\sigma(|x|)}$.
2. *If $x \notin L$, then* $\Pr[M(x)\ accepts] \leq 2^{-\sigma(|x|)}/2$.

The complexity class SBQP *is defined analogously, wherein the classical algorithm is replaced with a quantum algorithm.*

A classical (respectively, quantum) algorithm that satisfies the above promise for a particular language will be referred to as an SBP (respectively, SBQP) algorithm throughout. Using these definitions, a query complexity relation between QMA protocols and SBQP algorithms follows from the procedure of Marriott and Watrous [36], which shows that one can exponentially improve the soundness and completeness errors of a QMA protocol without increasing the witness size. This relationship is now standard; see for example [36, Remark 6] or [50, Proposition 4.2] for a proof of the following lemma:

Lemma 17. *Suppose there is a* QMA *protocol for some problem that makes T queries and receives an m-qubit witness. Then there is a quantum query algorithm Q for the same problem that makes $O(mT)$ queries, and satisfies the following:*

1. *If $x \in L$, then* $\Pr[Q(x)\ accepts] \geq 2^{-m}$.
2. *If $x \notin L$, then* $\Pr[Q(x)\ accepts] \leq 2^{-10m}$.

3 QMA complexity of approximate counting

This section establishes an optimal lower bound on the QMA complexity of approximate counting. We first lower bound the SBQP complexity of the $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ problem (Theorem 3). This implies a QMA lower bound for $\mathsf{ApxCount}_{N,w}$ via Lemma 17, but it is not quantitatively optimal. We prove the optimal QMA lower bound (Theorem 2) via Lemma 19, which leverages additional properties of the SBQP protocol derived via Lemma 17 from any QMA protocol with small witness length. Finally, Corollary 20 describes new oracle separations that are immediate consequences of Theorem 2 and Theorem 3.

3.1 Lower bound for SBQP algorithms

Our lower bound on the SBQP complexity of $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ hinges on the following theorem. The theorem uses Laurent polynomials to prove a degree lower bound for bivariate polynomials that satisfy an upper bound on an "L"-shaped pair of rectangles and a lower bound at a nearby point:

Theorem 18. *Let $0 < w < 32w < N$ and $M \geq 1$. Let $R_1 = [4w, N] \times [0, w/2]$ and $R_2 = [0, w/2] \times [4w, N]$ be disjoint rectangles in the plane, and let $L = R_1 \cup R_2$. Let $p(x, y)$ be a real polynomial of degree d with the following properties:*

1. *$p(4w, 4w) \geq 1.5 \cdot M$.*

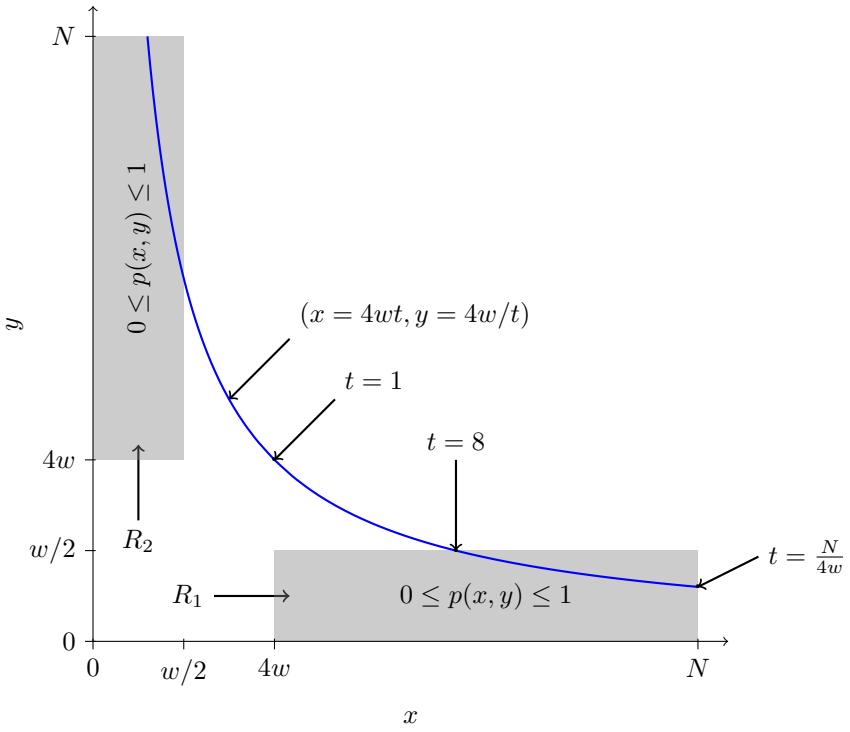


Figure 3: Diagram of Theorem 18 (not drawn to scale).

2. $0 \le p(x,y) \le 1$ *for all* $(x,y) \in L$.

Then $d = \Omega(\sqrt{N/w} \cdot \log M)$.

Proof. Observe that if $p(x,y)$ satisfies the statement of the theorem, then so does $p(y,x)$. This is because the constraints in the statement of the theorem are symmetric in x and y (in particular, because R_1 and R_2 are mirror images of one another along the line $x = y$; see Figure 3). As a result, we may assume without loss of generality that p is symmetric, i.e., $p(x,y) = p(y,x)$. Else, we may replace p by $\frac{p(x,y)+p(y,x)}{2}$ because the set of polynomials that satisfy the inequalities in the statement of the theorem are closed under convex combinations.

Consider the hyperbolic parametric curve $(x = 4wt, y = 4w/t)$ as it passes through R_1 (see Figure 3). We can view the restriction of $p(x,y)$ to this curve as a Laurent polynomial $\ell(t) = p(4wt, 4w/t)$ of positive and negative degree d. The bound of $p(x,y)$ on all of R_1 implies that $|\ell(t)| \le 1$ when $t \in [8, \frac{N}{4w}]$ and that $\ell(1) \ge 1.5$ (see Figure 3). Moreover, the condition that $p(x,y)$ is symmetric implies that $\ell(t) = \ell(1/t)$.

By Lemma 14 for symmetric Laurent polynomials, $\ell(t)$ can be viewed as a degree d polynomial $q(t+1/t)$. Under the transformation $s = t+1/t$, q satisfies $|q(s)| \le 1$ for $s \in [8+1/8, \frac{N}{4w} + \frac{4w}{N}]$ and $q(2) \ge 1.5M$. Note that the length of the interval $[8+1/8, \frac{N}{4w} + \frac{4w}{N}]$ is $\Theta(N/w)$ because $w < N$. By an appropriate affine transformation of q, we can conclude from Lemma 7 with $\mu = \Theta(w/N)$ that $d = \Omega(\sqrt{N/w} \cdot \log M)$. □

Why is Theorem 18 useful? One may be tempted to apply this theorem directly to the polynomial $p(x, y)$ obtained in Lemma 13 to conclude a degree lower bound (and thus a query complexity lower bound), as the "L"-shaped pair of rectangles $L = R_1 \cup R_2$ correspond to "no" instances of $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$, while $(4w, 4w)$ corresponds to a "yes" instance. However, even though $p(x, y)$ is bounded at lattice points in L, it need not be bounded along the entirety of L.[1]

To obtain a lower bound, we instead use the connection between the polynomials $p(x, y)$ and $q(s, t)$ from Lemma 13, and establish Theorem 3 from the introduction, restated for convenience:

Theorem 3. *Consider an* SBQP *algorithm for* $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ *that makes* T *queries to membership oracles for the two instances of* $\mathsf{ApxCount}_{N,w}$*. Then* $T = \Omega\left(\min\{w, \sqrt{N/w}\}\right)$.

Proof. Let $N > 32w$ (otherwise the theorem holds trivially). Since Q is an SBQP algorithm, we may suppose that Q accepts with probability at least 2α on a "yes" instance and with probability at most α on a "no" instance (note that α may be exponentially small in N). Take $p(x, y)$ and $q(s, t)$ to be the symmetrized bivariate polynomials of degree at most $2T$ defined in Lemma 13. Define $L' = ([0, w] \times [0, w]) \cup ([0, w] \times [2w, N]) \cup ([2w, N] \times [0, w])$. The conditions on the acceptance probability of Q for all S_0, S_1 that satisfy the $\mathsf{ApxCount}_{N,w}$ promise imply that $p(x, y)$ satisfies these corresponding conditions:

1. $1 \geq p(x, y) \geq 2\alpha$ for all $(x, y) \in ([2w, N] \times [2w, N]) \cap \mathbb{Z}^2$.
2. $0 \leq p(x, y) \leq \alpha$ for all $(x, y) \in L' \cap \mathbb{Z}^2$.

Our strategy is to show that if $T = o(w)$, then these conditions on p imply that the polynomial $q(s, t) \cdot \frac{0.9}{\alpha}$ satisfies the statement of Theorem 18 for all sufficiently large w. This in turn implies $T = \Omega(\sqrt{N/w})$. This allows us conclude that either $T = \Omega(w)$ or $T = \Omega(\sqrt{N/w})$, which proves the theorem.

Suppose $T = o(w)$, so that $p(x, y)$ and $q(s, t)$ both have degree $d = o(w)$. We begin by upper bounding $p(x, y)$ at the lattice points (x, y) outside of L'. We claim the following:

(a) $|p(x, y)| \leq \alpha \cdot a \cdot \exp(bd^2/w) \leq \alpha \cdot a \cdot \exp(bd)$ whenever $(x, y) \in L'$ and either x or y is an integer, where a and b are the constants from Lemma 10. This follows from Lemma 10 by fixing either x or y to be an integer and viewing the resulting restriction of $p(x, y)$ as a univariate polynomial in the other variable.

(b) $|p(x, y)| \leq \alpha \cdot a \cdot \exp(bd) \cdot \exp(2\sqrt{3}d) = \alpha \cdot a \cdot \exp((b + 2\sqrt{3})d)$ whenever $x \in [w, 2w]$, $y \in [0, w]$, and y is an integer. This follows Lemma 7: consider the univariate polynomial $p(\cdot, y)$ on the intervals $[0, w]$ and $[2w, 3w]$, where it is bounded by (a).

[1] One can nevertheless use this intuition to obtain a nontrivial (though suboptimal) lower bound by inspecting p alone. Using the Markov brothers' inequality (Lemma 6), if $\deg(p) = o(\sqrt{w})$, then the bounds on $p(x, y)$ at lattice points in L imply that $|p(x, y)| \leq 1 + o_w(1)$ for all $(x, y) \in L$. Thus, Theorem 18 applies if $\deg(p) = o(\sqrt{w})$, so overall we get a lower bound of $\Omega\left(\min\left\{\sqrt{w}, \sqrt{N/w}\right\}\right)$ for the SBQP query complexity of $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$. See arXiv:1902.02398 for details.

(c) $|p(x,y)| \le \alpha \cdot a \cdot \exp((b+2\sqrt{3})d) \cdot a \cdot \exp(bd^2/w) \le \alpha \cdot a^2 \cdot \exp((2b+2\sqrt{3})d)$ whenever $x \in [w, 2w]$ and $y \in [0, w]$. This follows from Lemma 10: consider the univariate polynomial $p(x, \cdot)$ on the interval $[0, w]$, where it is bounded at integer points by (b).

(d) $|p(x,y)| \le \alpha \cdot a^2 \cdot \exp((2b+2\sqrt{3})d) \cdot \exp(4dy/w) = \alpha \cdot a^2 \cdot \exp((2b+2\sqrt{3}+4y/w)d)$ whenever $x \in [0, N]$, $y \in [w+1, N]$, and x is an integer. This follows from Lemma 7: consider the univariate polynomial $p(x, \cdot)$ on the interval $[0, w]$, where it is bounded by (a) when $x \in [0, w]$ or $x \in [2w, N]$, or bounded by (c) when $x \in [w, 2w]$. By an affine shift, this corresponds to applying Lemma 7 with $\mu = 2y/w - 2$, with the observation that $\sqrt{2\mu + \mu^2} < \mu + 2$.

We now use this to upper bound $q(s,t)$ when $s \in [4w, N]$ and $t \in [0, w/2]$. Let X and Y be drawn from N-trial binomial distributions with means s and t, respectively, so that $q(s,t) = \mathbb{E}[p(X,Y)]$. Using the above bounds and basic probability, we have

$$0 \le q(s,t) = \mathbb{E}[p(X,Y)] \le \alpha \cdot \Bigg(\Pr[X \ge 2w, Y \le w] + \Pr[X \le 2w, Y \le w] \cdot a \cdot \exp\left(\left(b + 2\sqrt{3}\right) d\right) + \sum_{y=w+1}^{N} \Pr[Y = y] \cdot a^2 \cdot \exp\left(\left(2b + 2\sqrt{3} + 4y/w\right) d\right) \Bigg) \tag{27}$$

$$\le \alpha \cdot \Bigg(1 + \Pr[X \le 2w] \cdot a \cdot \exp\left(\left(b + 2\sqrt{3}\right) d\right) + \sum_{y=w+1}^{N} \Pr[Y \ge y] \cdot a^2 \cdot \exp\left(\left(2b + 2\sqrt{3} + 4y/w\right) d\right) \Bigg). \tag{28}$$

The probabilities above are easily bounded with a Chernoff bound:

$$q(s,t) = \mathbb{E}[p(X,Y)] \le \alpha \cdot \Bigg(1 + a \cdot \exp\left(\left(b + 2\sqrt{3}\right) d - w/2\right) + \sum_{y=w+1}^{N} a^2 \cdot \exp\left(\left(2b + 2\sqrt{3} + 4y/w\right) d - y/6\right) \Bigg). \tag{29}$$

Because a and b are universal constants from Lemma 10, when $d = o(w)$, the first exponential term becomes arbitrarily small for all sufficiently large w. Moreover, for all sufficiently large w, the remaining sum becomes bounded by a geometric sum. For some constant c, we have

$$\begin{aligned} \sum_{y=w+1}^{N} a^2 \cdot \exp\left(\left(2b + 2\sqrt{3} + 4y/w\right) d - y/6\right) &\le \sum_{y=w+1}^{\infty} c \cdot \exp\left(-y/12\right) \\ &\le \frac{c}{1 - \exp(-1/12)} \cdot \exp(-w/12) \\ &= o_w(1). \end{aligned}$$

Thus we conclude that $0 \le q(s,t) \le \alpha \cdot (1 + o_w(1))$ when $s \in [4w, N]$ and $t \in [0, w/2]$ (i.e., $(s,t) \in R_1$ in the statement of Theorem 18). By symmetry, we

can conclude the same bound when $s \in [0, w/2]$ and $t \in [4w, N]$ (i.e., $(s,t) \in R_2$ in the statement of Theorem 18).

Now, we lower bound $q(4w, 4w)$. Let X and Y be drawn from independent N-trial binomial distributions with mean $4w$, so that $q(4w, 4w) = \mathbb{E}[p(X,Y)]$. Then we have

$$\begin{aligned}\mathbb{E}[p(X,Y)] &\geq 2\alpha \cdot \Pr[X \geq 2w, Y \geq 2w] \\ &\geq 2\alpha \cdot (1 - \Pr[X \leq 2w] - \Pr[Y \leq 2w]) \\ &\geq 2\alpha \cdot (1 - 2\exp(-w/2)) \\ &\geq 2\alpha \cdot (1 - o_w(1))\end{aligned}$$

We conclude that $q(s,t) \cdot \frac{0.9}{\alpha}$ satisfies the statement of Theorem 18 (with $M = 1$) for all sufficiently large w. □

We remark that this lower bound is tight, i.e., there exists an SBQP algorithm that makes $O\left(\min\left\{w, \sqrt{N/w}\right\}\right)$ queries. The $O(\sqrt{N/w})$ upper bound follows from the BQP algorithm of Brassard, Høyer, and Tapp [14]. The $O(w)$ upper bound is in fact an SBP upper bound with the following algorithmic interpretation: first, guess $w+1$ items randomly from each of S_0 and S_1. Then, verify using the membership oracle that the first $w+1$ items all belong to S_0 and that the latter $w+1$ items all belong to S_1, accepting if and only if this is the case. Clearly, this accepts with nonzero probability if and only if $|S_0| \geq w+1$ and $|S_1| \geq w+1$.

3.2 Lower bound for QMA

In this section, we establish the optimal QMA lower bound (Theorem 2). We begin by quantitatively improving the SBQP lower bound for $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ of Theorem 3, under the stronger assumption that the parameter α in the SBQP protocol is not smaller than 2^{-w}. (In addition to a stronger conclusion, this assumption also permits a considerably simpler analysis than was required to prove Theorem 3).

Lemma 19. *Consider any quantum query algorithm $Q^{\mathcal{O}_{S_0}, \mathcal{O}_{S_1}}$ for $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ that makes T queries to the membership oracles $\mathcal{O}_{S_0}$ and $\mathcal{O}_{S_1}$ for the two instances of $\mathsf{ApxCount}_{N,w}$ and satisfies the following. For some $m = o(w)$, $\alpha = 2^{-m}$, and $M \in [1, \alpha^{-1}]$:*

1. *If $x \in L$, then $\Pr[Q(x) \text{ accepts}] \geq \alpha$.*
2. *If $x \notin L$, then $\Pr[Q(x) \text{ accepts}] \leq \alpha/(2M)$.*

Then $T = \Omega\left(\sqrt{N/w} \cdot \log M\right)$

Proof. As in the proof of Theorem 3, define $L' = ([0,w] \times [0,w]) \cup ([0,w] \times [2w, N]) \cup ([2w, N] \times [0,w])$, and take $p(x,y)$ and $q(s,t)$ to be the symmetrized bivariate polynomials of degree at most $2T$ defined in Lemma 13. $p(x,y)$ satisfies the following properties.

(a) $1 \geq p(x,y) \geq \alpha$ for all $(x,y) \in ([2w, N] \times [2w, N]) \cap \mathbb{Z}^2$.

(b) $0 \leq p(x,y) \leq \alpha/(1.5M)$ for all $(x,y) \in L' \cap \mathbb{Z}^2$.

(c) $0 \leq p(x,y) \leq 1$ for all $(x,y) \in ([0,N] \times [0,N]) \cap \mathbb{Z}^2$.

We use these properties to upper bound $q(s,t)$ when $s \in [4w, N]$ and $t \in [0, w/2]$. Let X and Y be drawn from N-trial binomial distributions with means s and t, respectively, so that $q(s,t) = \mathbb{E}[p(X,Y)]$. Using the above bounds and basic probability, we have

$$\begin{aligned} 0 \leq q(s,t) = \mathbb{E}[p(X,Y)] &\leq \alpha/(2M) \Pr[X \geq 2w, Y \leq w] + (1 - \Pr[X \geq 2w, Y \leq w]) \\ &\leq \alpha/(2M) + 2^{-\Omega(w)} \leq (1 + o(1))\alpha/(2M) \end{aligned}$$

Here, the first inequality holds by Properties (a)-(c) above, while the second follows from a Chernoff Bound, and the third holds because $\alpha/(2M) \geq 2^{-o(w)}$.

Thus we conclude that $0 \leq q(s,t) \leq \alpha/(2M) \cdot (1 + o_w(1))$ when $s \in [4w, N]$ and $t \in [0, w/2]$ (i.e., $(s,t) \in R_1$ in the statement of Theorem 18). By symmetry, we can conclude the same bound when $s \in [0, w/2]$ and $t \in [4w, N]$ (i.e., $(s,t) \in R_2$ in the statement of Theorem 18).

Now, we lower bound $q(4w, 4w)$. Let X and Y be drawn from independent N-trial binomial distributions with mean $4w$, so that $q(4w,4w) = \mathbb{E}[p(X,Y)]$. Then we have

$$\begin{aligned} \mathbb{E}[p(X,Y)] &\geq \alpha \cdot \Pr[X \geq 2w, Y \geq 2w] \\ &\geq \alpha \cdot (1 - \Pr[X \leq 2w] - \Pr[Y \leq 2w]) \\ &\geq \alpha \cdot (1 - 2\exp(-w/2)) \\ &\geq \alpha \cdot (1 - o_w(1)) \end{aligned}$$

We conclude that $q(s,t) \cdot \frac{1.8M}{\alpha}$ satisfies the statement of Theorem 18 for all sufficiently large w. Hence, $T = \Omega\left(\sqrt{N/w} \cdot \log M\right)$ as claimed. □

We now establish Theorem 2 from the introduction, which quantitatively lower bounds the QMA complexity of $\mathsf{ApxCount}_{N,w}$. The analysis exploits two key properties of the SBQP protocols that result from applying Lemma 17 to a QMA protocol with witness length m: (1) the parameter α of the SBQP protocol is not too small (at least 2^{-m}) and (2) the multiplicative gap between acceptance probabilities when $f(x) = 0$ vs. $f(x) = 1$ is at least 2^m, which may be much greater than 2.

Theorem 2. *Consider a* QMA *protocol that solves* $\mathsf{ApxCount}_{N,w}$. *If the protocol receives a quantum witness of length m, and makes T queries to the membership oracle for S, then either $m = \Omega(w)$ or $T = \Omega\big(\sqrt{N/w}\big)$.*

Proof. Consider a QMA protocol for $\mathsf{ApxCount}_{N,w}$ with witness size m and query cost T. If $m = \Omega(w)$, the theorem is vacuous, so suppose that $m = o(w)$. Running the verifier, Arthur, a constant number of times with fresh witnesses to reduce the soundness and completeness errors, one obtains a verifier with soundness and completeness errors $1/6$ that receives an $O(m)$-length witness and makes $O(T)$ queries. Repeating twice with two oracles and computing the AND, one obtains a QMA verifier $V'^{\mathcal{O}_{S_0}, \mathcal{O}_{S_1}}$ for $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ with soundness and completeness errors $1/3$ that receives an $O(m)$-length witness and makes $O(T)$ queries. Applying Lemma 17 to V', there exists a quantum query algorithm $Q^{\mathcal{O}_{S_0}, \mathcal{O}_{S_1}}$ for $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ that makes $O(m \cdot T)$ queries

and satisfies the hypothesis of Lemma 19 with $M = 2^{-\Theta(m)}$. Theorem 3 tells us that $m \cdot T = \Omega\left(\sqrt{N/w} \cdot m\right)$. Equivalently, $T = \Omega\left(\sqrt{N/w}\right)$. □

Theorem 3 also implies several oracle separations:

Corollary 20. *There exists an oracle A and a pair of languages L_0, L_1 such that:*

1. $L_0, L_1 \in \mathsf{SBP}^A$
2. $L_0 \cap L_1 \notin \mathsf{SBQP}^A$.
3. $\mathsf{SBP}^A \not\subset \mathsf{QMA}^A$.

Proof. For an arbitrary function $A : \{0,1\}^* \to \{0,1\}$ and $i \in \{0,1\}$, define $A_i^n = \{x \in \{0,1\}^n : A(i,x) = 1\}$. Define the unary language $L_i^A = \{1^n : |A_i^n| \geq 2^{n/2}\}$. Observe that as long as A satisfies the promise $|A_i^n| \geq 2^{n/2}$ or $|A_i^n| \leq 2^{n/2-1}$ for all $n \in \mathbb{N}$, then $L_i^A \in \mathsf{SBP}^A$. Intuitively, the oracles A that satisfy this promise encode a pair of $\mathsf{ApxCount}_{N,w}$ instances $|A_0^n|$ and $|A_1^n|$ for every $n \in \mathbb{N}$ where $N = 2^n$ and $w = 2^{n/2-1}$.

Theorem 3 tells us that an SBQP algorithm Q that makes $o(2^{n/4})$ queries fails to solve $\mathsf{AND}_2 \circ \mathsf{ApxCount}_{N,w}$ on *some* pair (S_0, S_1) that satisfies the promise. Thus, one can construct an A such that $L_0, L_1 \in \mathsf{SBP}^A$ and $L_0 \cap L_1 \notin \mathsf{SBQP}^A$, by choosing (A_0^n, A_1^n) so as to diagonalize against all SBQP algorithms.

Because QMA^A is closed under intersection for any oracle A, and because $\mathsf{QMA}^A \subseteq \mathsf{SBQP}^A$ for any oracle A, it must be the case that either $L_0 \notin \mathsf{QMA}^A$ or $L_1 \notin \mathsf{QMA}^A$. □

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